MOBILE COMPUTING LABORATORY
FIGMA DESIGN QUESTIONS

1. Use Figma to Create an Interactive Diagram Illustrating the Components of a Basic Cellular System

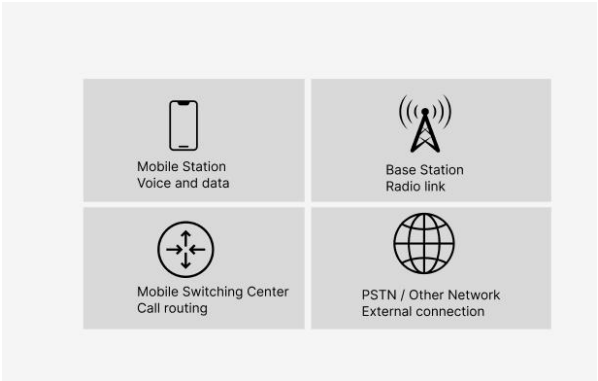
AIM

To create an interactive diagram in Figma showing the main components of a basic cellular communication system and how they communicate with one another.

PROCEDURE

1. Open https://www.figma.com/ and create a new design file.
2. Press F to create a frame suitable for a diagram. Rename it "Basic Cellular System".
3. Use rectangles to represent the Mobile Station (MS), Base Transceiver Station (BTS), Base Station Controller (BSC), Mobile Switching Centre (MSC), and Public Switched Telephone Network (PSTN) or another mobile network.
4. Arrange the components in communication order: MS → BTS → BSC → MSC → external network or destination subscriber.
5. Use arrows or connectors to show the direction of communication. Add labels such as "Radio link", "Base-station control", and "Call switching".
6. Use different colors for the mobile device, radio access network, switching network, and external network.
7. Open the Prototype tab and connect clickable components or information buttons to detail frames or popovers represented by separate frames.
8. Click Present to test the interactions. Save or export the diagram as a PDF or image if required.

DESIGN



- Mobile Station (MS): the user's mobile phone that originates or receives calls.
- Base Transceiver Station (BTS): provides the radio link between the mobile station and the network.
- Base Station Controller (BSC): manages radio resources and controls multiple BTSs in a GSM-style network.
- Mobile Switching Centre (MSC): switches calls and coordinates mobility-related functions.

- External network / destination: connects the call to another subscriber or a public telephone network.

RESULT

An interactive diagram of a basic cellular system was designed in Figma. It illustrates the main network components and the communication path from the mobile station through the radio access and switching network to the destination.

2. Design a User Interface Prototype for a Mobile Station Using Figma, Considering Connectivity and Convenience

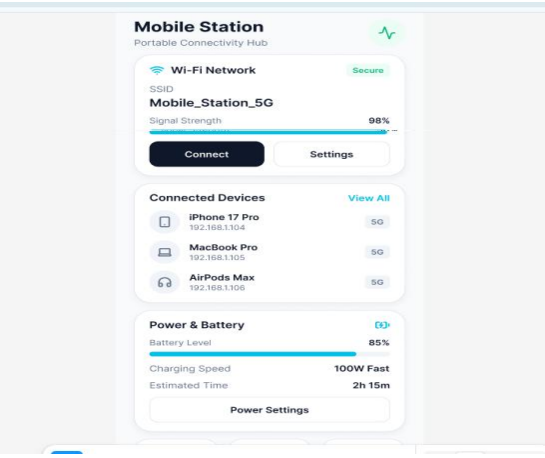
AIM

To design a user-friendly mobile station interface in Figma that provides quick access to connectivity status, Wi-Fi, mobile data, data usage, and common settings.

PROCEDURE

9. Open Figma and create a new design file.
10. Press F and choose a phone preset or set the frame to approximately 390 × 844 pixels. Name it "Home Screen".
11. Choose a consistent color palette, such as blue, white, and light grey, and use a readable font such as Inter.
12. Create a header titled "Mobile Station" and a network status card showing connected/disconnected status and signal strength.
13. Add Wi-Fi and Mobile Data cards, a data-usage panel with a progress bar, and a "Manage Connections" button.
14. Create additional frames named "Connectivity", "Data Usage", and "Settings". Include Wi-Fi networks, mobile data status, usage details, notifications, and accessibility options.
15. Open the Prototype tab and connect the buttons and bottom navigation icons to the corresponding frames using "On tap → Navigate to".
16. Click Present to test navigation. Remember that toggles in a Figma prototype simulate the interface; they do not control the phone's real network settings.

DESIGN



- Home Screen: displays network status, signal strength, Wi-Fi, mobile data, battery indicator, and data usage.
- Connectivity Screen: shows Wi-Fi and mobile data controls and a list of available networks.
- Data Usage Screen: displays the data consumed, remaining allowance, and a usage progress bar.
- Settings Screen: includes network preferences, notifications, accessibility, and help.
- Convenience features: clear labels, large touch targets, consistent navigation, and a message for offline or weak-network conditions.

RESULT

A mobile station UI prototype was designed in Figma with connectivity controls, data-usage information, and simple navigation. The prototype demonstrates how common mobile network functions can be presented in a clear and convenient interface.

3. Develop a Flowchart in Figma Depicting the Operation of a Network Switching Subsystem in a Mobile Cellular System

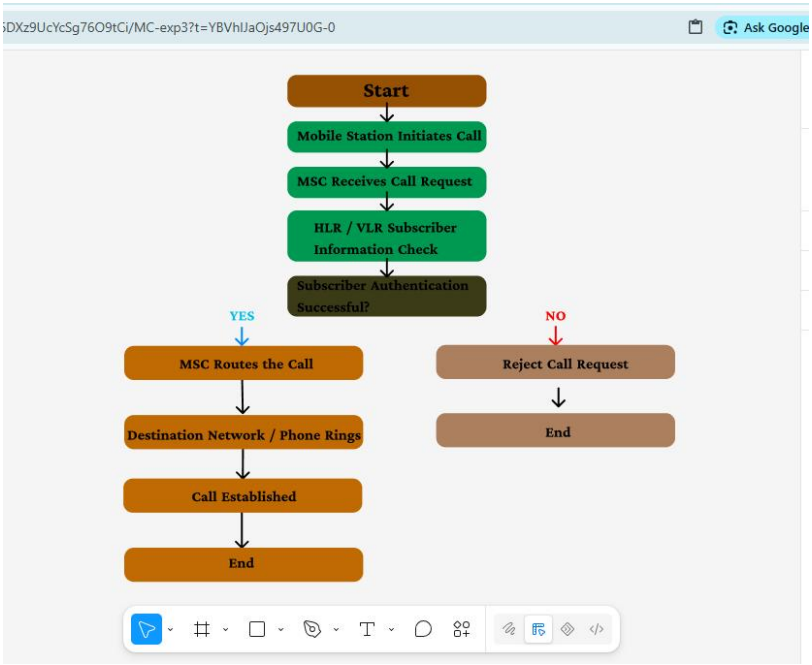
AIM

To create a flowchart in Figma that explains the operation of the Network Switching Subsystem (NSS), including subscriber information checks, authentication, call routing, and call termination.

PROCEDURE

17. Open Figma and create a new design file with a frame named “Network Switching Subsystem Flowchart”.
18. Use oval shapes for Start and End, rectangles for process steps, a diamond for the authentication decision, and arrows for flow direction.
19. Add the following process blocks: Start; Mobile Station initiates a call; MSC receives the request; HLR/VLR subscriber information is checked; subscriber authentication is performed.
20. Create a decision diamond labelled “Authentication successful?”.
21. From the Yes branch, connect to “MSC routes the call”, then “Destination phone rings”, “Call established”, and “End”.
22. From the No branch, connect to “Reject call request” and then “End”. Label the branches Yes and No.
23. Use different colors for process boxes, decision boxes, and start/end shapes. Keep the flow top-to-bottom and avoid crossing arrows.
24. Use Present to review the diagram, then export or share it for submission.

DESIGN



- MSC (Mobile Switching Centre): handles call switching and routing.
- HLR (Home Location Register): stores permanent subscriber information.
- VLR (Visitor Location Register): stores temporary subscriber and location information.
- AuC (Authentication Centre): supports verification of subscriber identity.
- EIR (Equipment Identity Register): can check equipment identity when the network uses this function.
- Flow: Start → Call request → Subscriber data check → Authentication decision → Route call if accepted, or reject if authentication fails.

RESULT

A flowchart describing the main call-processing steps of the Network Switching Subsystem was designed in Figma. It shows subscriber information retrieval, authentication, call routing, and call rejection when authentication fails.

4. Create a Visual Representation in Figma Highlighting the Key Elements of Mobile Network Protocol Stacks

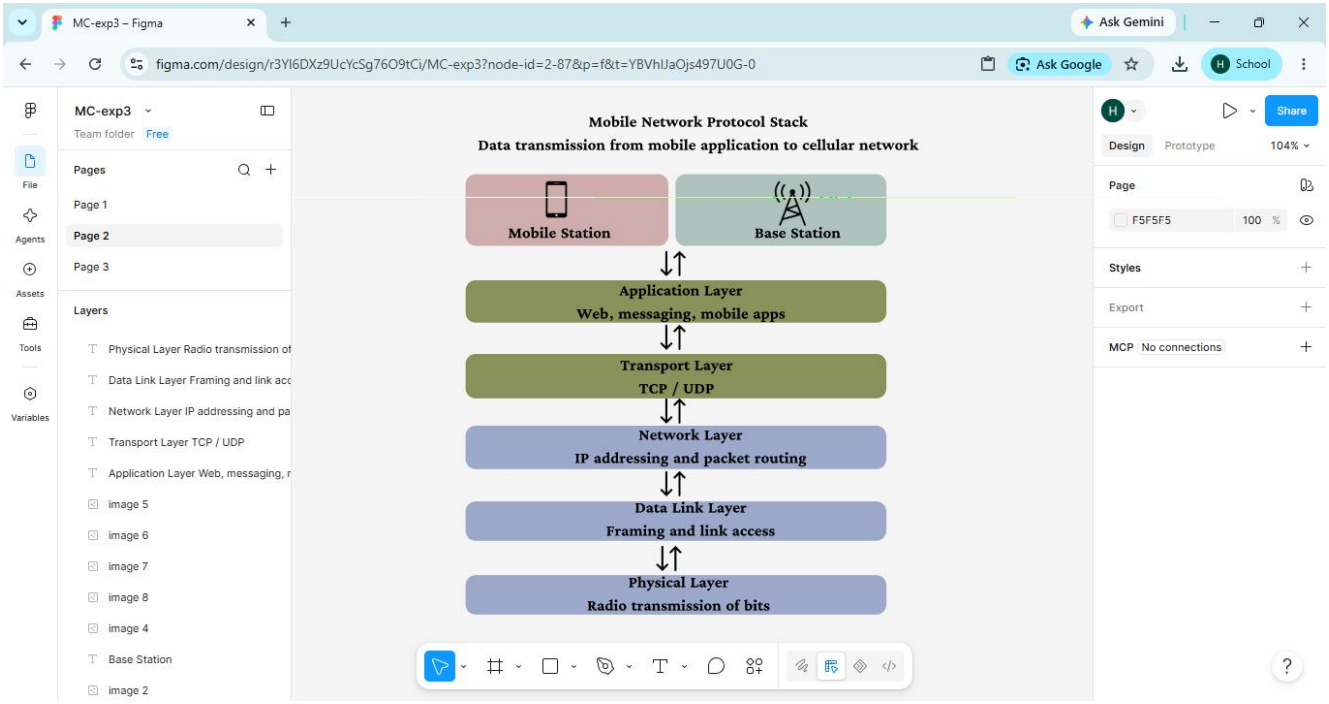
AIM

To design a visual representation in Figma showing key protocol layers and their functions in mobile communication, and how data travels between a mobile station and the cellular network.

PROCEDURE

25. Open Figma and create a new design file. Add a frame titled “Mobile Network Protocol Stack”.
26. Use five vertically stacked rectangles to represent the Application, Transport, Network, Data Link, and Physical layers.
27. Add short descriptions and protocol examples: applications; TCP/UDP; IP; link framing and access; and radio transmission of bits.
28. Place a Mobile Station icon on one side and a Base Station/Cellular Network icon on the other side.
29. Use arrows between layers to show data moving down the stack for transmission and up the stack at the receiving side.
30. Apply different colors to the layers and use consistent spacing, text sizes, and alignment.
31. Add a note that actual cellular protocol stacks differ by technology and interface. The five-layer model is a simplified conceptual view.
32. Use Present to inspect the layout, then export the frame as an image or PDF.

DESIGN



- Application Layer: provides user-facing services such as web browsing and messaging.
- Transport Layer: TCP and UDP provide transport services for application data.
- Network Layer: IP addressing and packet routing support communication across networks.

- Data Link Layer: manages link-level framing and access to the communication medium.
- Physical Layer: transmits bits over the radio interface.
- Use arrows to indicate encapsulation and transmission from the application toward the physical medium; reverse the direction at the receiving end.

RESULT

A visual representation of mobile network protocol layers was designed in Figma. It highlights the main layer functions and provides a simplified explanation of how data is passed through a protocol stack during mobile communication.

5. Design a Wireframe for a Mobile Application That Incorporates Localization Features Using Figma

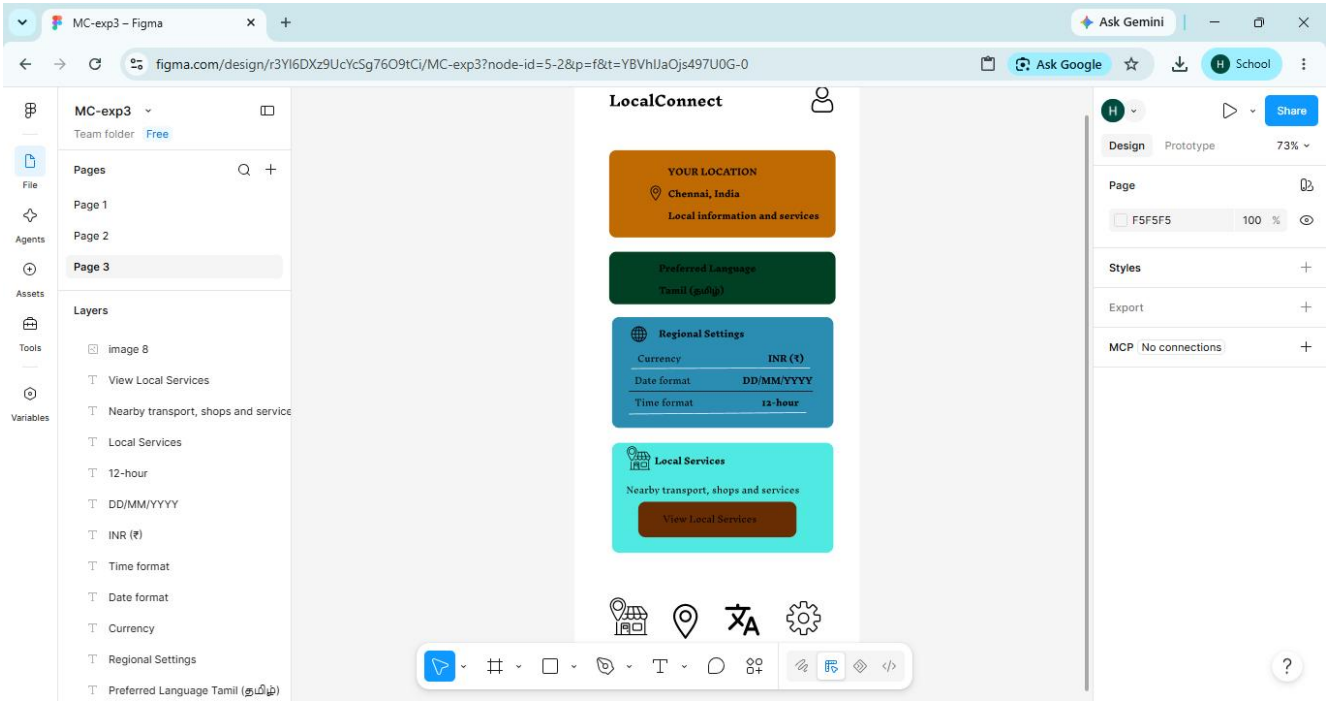
AIM

To design a mobile application wireframe in Figma that incorporates localization features such as language selection, regional currency, date and time formats, and location-based information.

PROCEDURE

33. Open Figma and create a new design file. Press F and select a mobile frame preset or use approximately 390 × 844 pixels.
34. Add an application title such as “LocalConnect” at the top of the screen.
35. Create wireframe sections using outlined rectangles for Location, Preferred Language, Regional Settings, and Local Services.
36. Add a language dropdown with example options such as English and Tamil.
37. Add regional settings fields for country/region, currency (for example INR ₹), date format (DD/MM/YYYY), and time format (12-hour or 24-hour).
38. Add a local services section for nearby transport, shops, or other relevant services. Treat the location as sample content unless real location integration is implemented.
39. Create separate frames for Language Selection and Regional Settings. Connect buttons and dropdown-like controls to these frames in the Prototype tab.
40. Use simple grey outlines, readable labels, and consistent spacing to keep the design in wireframe style. Test navigation in Present mode.

DESIGN



- Home Screen: application title, current region, and shortcuts to localized content.
- Language Selection: allows users to choose their preferred display language.
- Regional Settings: allows selection of country/region, currency, date format, and time format.
- Location-Based Services: presents local transport, shops, or regional information based on the selected or permitted location.
- Usability considerations: clearly labelled controls, readable text, and an option to change language or region later.

RESULT

A mobile application wireframe with localization features was designed in Figma. It demonstrates language selection, regional currency and date/time settings, and location-based services to support a more accessible and locally relevant user experience.